

506021334 Stats102A HW1

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Sourcing the R file(file should be within the working directory)

```
source("./506021334_stats102a_hw1.R")
```

```
# in class test cases  
print(gcd(48, 18L)) # answer: 6
```

```
## [1] 6
```

```
print(gcd(180, 25L)) # answer: 5
```

```
## [1] 5
```

```
# in class test cases  
gcdi(c(16, 32, -58)) # 2
```

```
## [1] 2
```

```
gcdi(c(4789, 6123, 199)) # 1
```

```
## [1] 1
```

```
# in class test cases  
lcm(4, 6)
```

```
## Warning in lcm(4, 6): lcm: Inputs are not integers, coercing x and y using  
## as.integer...
```

```
## [1] 12
```

```
# in class test cases  
add_2_frac(1, 2, 1, 3) # 5/6
```

```
## Warning in add_2_frac(1, 2, 1, 3): add_2_frac: Found inputs that are not  
## integers, coercing n1, d1, n2, d2 using as.integer...
```

```
## $num
## [1] 5
##
## $denom
## [1] 6
```

```
add_2_frac(1, 11, 1, 5) # list(num = 16, denom = 55)
```

```
## Warning in add_2_frac(1, 11, 1, 5): add_2_frac: Found inputs that are not
## integers, coercing n1, d1, n2, d2 using as.integer...
```

```
## $num
## [1] 16
##
## $denom
## [1] 55
```

```
add_2_frac(1, 11, -1, 5) # list(num = -6, denom = 55)
```

```
## Warning in add_2_frac(1, 11, -1, 5): add_2_frac: Found inputs that are not
## integers, coercing n1, d1, n2, d2 using as.integer...
```

```
## $num
## [1] -6
##
## $denom
## [1] 55
```

```
# in class test cases
is_prime(c(9, 7)) # [False True]
```

```
## Warning in is_prime(c(9, 7)): is_prime: Input x is not an integer vector,
## coercing all values of x using as.integer...
```

```
## [1] TRUE TRUE
```

```
is_prime(1)
```

```
## Warning in is_prime(1): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
```

```
## [1] FALSE
```

```
is_prime(c(1, 3, 5))
```

```
## Warning in is_prime(c(1, 3, 5)): is_prime: Input x is not an integer vector,
## coercing all values of x using as.integer...
```

```
## [1] FALSE TRUE TRUE
```

```
is_prime(c(1, 2, 4, 7, 9, 10))
```

```
## Warning in is_prime(c(1, 2, 4, 7, 9, 10)): is_prime: Input x is not an integer  
## vector, coercing all values of x using as.integer...
```

```
## [1] FALSE TRUE FALSE TRUE TRUE FALSE
```

```
is_prime(c(29, 97, 143, 169, 221, 323, 377, 493, 541, 599))
```

```
## Warning in is_prime(c(29, 97, 143, 169, 221, 323, 377, 493, 541, 599)):  
## is_prime: Input x is not an integer vector, coercing all values of x using  
## as.integer...
```

```
## [1] TRUE TRUE FALSE FALSE FALSE FALSE FALSE TRUE TRUE
```

```
# in class test cases
```

```
get_factors(1920) # list(primes = [2 3 5], exponents = [7 1 1])
```

```
## Warning in get_factors(1920): get_factors: Input x is not an integer, coercing  
## x using as.integer...
```

```
## [1] 960  
## [1] 480  
## [1] 240  
## [1] 120  
## [1] 60  
## [1] 30  
## [1] 15
```

```
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...
```

```
## [1] 5
```

```
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...
```

```
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...
```

```
## [1] 1
```

```
## $primes  
## [1] 2 3 5  
##  
## $exponents  
## [1] 7 1 1
```

```
get_factors(1.92) # appropriate error handling
```

```
## Warning in get_factors(1.92): get_factors: Input x is not an integer, coercing  
## x using as.integer...
```

```
## $primes  
## NULL  
##  
## $exponents  
## NULL
```

```
x <- sample(x = 1e4, size = 1)  
y <- get_factors(x)
```

```
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
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## all values of x using as.integer...  
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## all values of x using as.integer...  
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## all values of x using as.integer...  
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## all values of x using as.integer...  
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## all values of x using as.integer...  
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## all values of x using as.integer...  
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## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...  
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing  
## all values of x using as.integer...
```

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```
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
```

```
## [1] 1
```

```
this_works <- prod(y$primes^y$exponents) == x & all(is_prime(y$primes))
```

```
## Warning in is_prime(y$primes): is_prime: Input x is not an integer vector,
## coercing all values of x using as.integer...
```

```
if (this_works) {
  print("passed test case!")
} else {print("false!")}
```

```
## [1] "passed test case!"
```

```
print(get_factors(1532512))
```

```
## Warning in get_factors(1532512): get_factors: Input x is not an integer,
## coercing x using as.integer...
```

```
## [1] 766256
## [1] 383128
## [1] 191564
## [1] 95782
## [1] 47891
```

```
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
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## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
## Warning in is_prime(incr): is_prime: Input x is not an integer vector, coercing
## all values of x using as.integer...
```

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```
## all values of x using as.integer...
```

```
## [1] 1
```

```
## $primes
```

```
## [1] 2 83 577
```

```
##
```

```
## $exponents
```

```
## [1] 5 1 1
```